

Protein Engineering Assignment: Stepwise guide to stabilize a helical protein

(Total Marks: 40)

Introductory class

- Step 1: Demonstration to install Windows Subsystem Linux (or miniconda)
- Step 2: Demonstration to install PyMOL and Discovery Studio for molecular visualization.
- Step 3: AGADIR installation and use (or pyAGADIR for miniconda)

Module 1: Identification of surface residues in a helix for mutation (Timeline – Complete by 11 March 2026)

- Step 1: Download the PDB file of AvrPto protein (PDB 1D – 1R5E) from Protein Data Bank.
- Step 2: Open the file in PyMOL, identify the surface residues in helix H1.
- Step 3: Identify the molecular interactions of the residues targeted for mutation with BIOVIA Discovery Studio Visualizer
- Your group should send a list of surface residues in helix H1, which you will mutate. Send this list by email to Dr. Soumya De (somde@iitkgp.ac.in) and Dr. Dibyendu Samanta (dibyendu.samanta@iitkgp.ac.in) by 11th March midnight. Write a report with appropriate figures and rationale for selection of the surface residues.
- 5 marks.

Module 2: Designing helix-stabilizing mutations using AGADIR (Timeline – Complete by 18 March 2026)

- Step 4: Redesign the sequence of Helix 1 using AGADIR to improve alpha-helical propensity.
- Step 5: You may also use other tools like HeliQuest, Chou-Fasman, etc to characterize a helix.
- Your group should send a list of potential mutations in helix H1 by 18th March midnight. Write a report with appropriate figures and rationale for the observed helical propensities of the designed mutants.
- 5 marks

Module 3: Design of 3D models of the mutant proteins using AlphaFold3 (Timeline – Complete by 25 March 2026)

- Step 5: Generate 3D models of the mutant proteins using AlphaFold server.
- Submit the 3D models of your structures. Write a report with appropriate figures.
- 5 marks

Module 4: Determination of protein stability (Timeline – Complete by 01 April 2026)

- Step 7: Using DynaMut2, istable2.0, and other webserver to assess the stability of the mutants.
- Step 8: Brief describe the experimental methods to study protein stabilization.
- Write a report with appropriate figures.
- 5 marks

Final Presentations will be held on 16th and 17th April. All members in a group have to be present during presentation and answer questions. 20 marks.